# Nischal Ashok

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## RESEARCH INTERESTS

Machine and Deep Learning, Natural Language Processing, Multi-modal Analysis, Explainability and Robustness

## EDUCATION BACKGROUND

**Indian Institute of Technology (IIT) Patna** — **July 2018-May 2022**
*Bachelor of Technology (B.Tech), Computer Science and Engineering* — ***CGPA: 9.20/10***

## PUBLICATIONS

* denotes equal contribution

- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"A Multi-task Learning Approach for Fake News Detection: Novelty, Emotion, and Sentiment Lend a Helping Hand"**, International Joint Conference on Neural Networks, **IJCNN-2021**, **[Accepted]**
- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"Misinformation detection using multi-task learning with mutual learning for novelty detection and emotion recognition"**, Journal of Information Processing and Management, Elsevier, **IPM-2021**, (IF:4.787) [Paper] [Code]
- Nischal Ashok, Jan Philip Wahle, Terry Ruas, Norman Meuschke, Tirthankar Ghosal and Bela Gipp, **"Empirical Studies on Health Misinformation Detection"**, [under review]
- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"Multi-Modal Dataset for Fake News Detection"**, [under review]
- Nischal Ashok*, Rina Kumari*, Tirthankar Ghoshal and Asif Ekbal, **"What the fake? Probing Misinformation Detection Standing on the Shoulder of Novelty and Emotion"**, Journal of Information Processing and Management, **IPM**, Elsevier (IF:4.787) [under review]
- Nischal Ashok, Venkatesh Babu R, **"Exploring Adversarial Robustness in Deep Neural Networks"**, (Technical Report) Indian Academy of Sciences Research Portal **[Accepted]**

## EXPERIENCE

**Intern, IBM Research AI** — **IBM, Bangalore, India**
*Research Managers: • Nitin Gupta • Hima Patel* — *May 2021-Present*

- **Project: Explainable Data Quality for Data Disambiguation**
- Building a novel self-explainable multi-task neural network architecture for detecting data issues for improving generalizability of programs for data transformation using programming by examples.

**Undergraduate Researcher, DKE Group** — **University of Wuppertal, Germany**
*Advisors: • Dr. Terry Ruas • Prof. Dr. Bela Gipp* — *Dec 2020-May 2021*

- **Project: Language Technologies for detecting health fake news**
- Performed and Analyzed Transfer Learning based NLP approaches on health-related fake-news datasets.
- Probed the efficacy of various neural language models with varying training schemes on domain-specific data.

**Undergraduate Researcher, AI-NLP-ML Lab** — **IIT-Patna, India**
*Advisors: • Prof. Asif Ekbal* — *July 2019-Present*

- **Present Project: Multi-modal Misinformation Detection**
- Probed misinformation detection through novelty detection, emotion recognition and sentiment analysis.
- Proposed a novel multi-tasking deep neural network architecture giving State-of-the-Art results on 4 well known datasets (ByteDance, FNC, Covid-Stance and FNID with 7.73%, 3.69%, 7.95% and 13.38% improvement)

**AI Research Intern, Video Analytics Laboratory** — **Indian Institute of Science, Bangalore, India**
*Advisor: • Prof. Venkatesh Babu* — *May 2020-Aug 2020*

- **Project: Unlabeled Data improves Adversarial Robustness**
- Experimented with semi-supervised learning algorithms to utilize unlabeled data from different distributions by proposing novel loss functions using cross-entropy coupled with KL-divergence
- Performed domain-adaptation using DE (Data-Enriching)-GANs between Cifar-100 and Cifar-10 datasets.
- Showed that unlabeled data from a different distribution is as competitive as the fully-supervised setting (within 2% error margin).

**AI Startup Winter Intern, TildeHat** — **Bangalore, India**
*Manager: • Dr. Om Deshmukh* — *Dec 2019-Jan 2020*

- **Project: Unsupervised Data Retrieval**
- Designed end-to-end module by converting pdf and doc files to HTML format and analyzed its DOM structure
- Implemented unsupervised algorithms and hierarchical classifier approaches for data-retrieval over 1000 documents

## PROJECTS

**IIT-Patna Academic Alexa** **IIT Patna**
*Guide: Prof. Pushpak Bhattacharya* *Jan 2020-May 2020*

- A sentiment-aware intelligent information retrieval based Chatbot for the Academic Portal of IIT-Patna.
- Devised a novel and effective way of converting natural language data into SQL queries using Dependency Tree Parsers, POS tagging, and NER techniques for information retrieval from the database
- Designed a custom chatbot pipeline which includes Multi-Channel CNN based intent classifier, sentiment analyzer, sentence similarity and feedback modules wrapped in wx-python GUI interface [Code] [Blog]

**Reddit-Drug Abuse Detection** **Data Analytics Lab, IIT Patna**
*Guide: Dr. Joydeep Chandra* *Jan 2020-May 2020*

- Investigated Drug Abuse on Social Media through Reddit Website by collecting and annotating data from r-opiates sub-reddit.
- Built a multi-head attention deep neural network for classifying comments as motivational or not. Further classification done on altruism, hope, good advice, bad advice, and universality. [Data+Code]

**Computer Science Academic Projects** **IIT-Patna**
*Guides: • Dr. Jimson Mathew • Dr. Samrat Mondal* *May 2020-May 2021*

- **Two Pass Assembler and Emulator** - Built an assembler-emulator system to convert the given assembly language code into a binary object file and then emulate/execute it. [Code]
- **Smart Lock System** - Built a software simulation of a smart lock system using two-step authentication of face recognition (using OpenCV) and pin lock system (implemented in Verilog). [Code]
- **HoneyPot based Authentication System** - Implemented Eugler's HoneyPot based Password Authentication System for providing additional security against traditional password cracking approaches. [Code]

## AWARDS and HONORS

- **Conference Grant Scholarships** - ICWSM 2021-AAAI, NAACL-2021
- **Summer Research Fellow** (FASTSF'2020) - Indian Academy of Sciences [India's foremost Science Academy] for pursuing research internship in Artificial Intelligence at IISc, Bangalore.
- **IIT-Patna Merit List** - Among top 10 CGPA holder in the IIT-Patna Computer Science class of 2022.
- **Fellowship (KVPY'2016)** - Young Scientist Incentive Plan - Government of India, for pursuing a career in research in pure sciences. Stood among top 0.2% among 200,000 students.
- **Academic Excellence Award** - Ranked 2nd among 150,000 students in 10th Standard public exam (ICSE'2016).

## SERVICE

- **Volunteer** - NAACL-2021
- **Head of ML and AI** - NJACK - Computer Science Club (IIT-Patna) - Introduced NJACK ML Workshop Series.
- **Departmental Lead** - DSC IIT-P - Associate Facilitator of Explore-ML Workshop IIT-P powered by Google-AI
- **Mentor** - Institute Student Mentorship Program (IIT-Patna) (2020-Present)
- **Mentor** - NJACK Winter of Code Open Source Programme'2019 - [ML Project Link]
- **Teaching** [Link] and **writing blogs** [Link] on ML and AI
- **Coordinator** - E-Cell - Introduced Entrepreneur-101 series and organized 4 sessions on entrepreneurship.
- **Social Worker** - National Service Scheme IIT-Patna - Led the rural entrepreneurship division.

## SKILLS

- **Programming Languages:** • C/C++ • Python • Java • Bash • Verilog • Assembly
- **Deep Learning Tools:** • Tensorflow • Keras • PyTorch • HuggingFace
- **Other tools:** • SQL • Docker • LaTeX • Git/GitHub